Native lane-flow equations: two admissibility checks

2026-10-20 — analytical review, no traffic rollout

Finding. Literal source receiving adjustment can force negative forward flow in a positive-density state. Separately, two pairwise lateral desires can exceed their donor's common sending budget. These checks concern the displayed equations, not an unobserved author implementation. They do not establish a replacement rule, calibrated validity or a gradient result.

1. Source rules and the exact interpretation checked

Use the notation of Nagalur Subraveti et al. [1], with $L_{i,l\to l'}$ abbreviating their realized lateral flow $lq_{il\to l'}$. Density k is in veh/km and all D, S, L, q are in veh/h. Their equations 6 and 13–17 give

$$\mathcal{P}_{i,l\to l'} = \max\left\{0, \frac{IK_{il} - K_{il'}}{K_{il} + K_{il'}}\right\}, \qquad D_{il} = \min\{u_{il}k_{il}, C_{il}\}, \tag{1}$$

$$S_{il} = \min\{C_{il}, w_{il}(k_{il}^{\text{jam}} - k_{il})\}, \quad \phi_{i,l\to l'} = D_{il}\mathcal{P}_{i,l\to l'}, \quad \theta_{il'} = \min\{1, (C_{il'} - S_{il'})/C_{il'}\}. \tag{2}$$

Here K is a convex weighted average of current and two downstream densities on that lane. For $\theta < 1$, the printed realized flow is $L = (1-\theta)\phi$. In both checks the targets have $S = C$, so $\theta = 0$ and $L = \phi$; the disputed special branch at $\theta = 1$ is never used.

For one lateral pair, source equations 20–22 print

$$\begin{aligned} D_{il}^{\text{mod}} &= D_{il} + L_{i,l'\to l} - L_{i,l\to l'}, \\ S_{il}^{\text{mod}} &= S_{il} + L_{i,l'\to l} - L_{i,l\to l'}, \quad q_{i-1,l} = \min\{D_{i-1,l}^{\text{mod}}, S_{il}^{\text{mod}}\}. \end{aligned} \tag{3}$$

The second line reindexes the source receiving cell from $i+1$ to i. It is not a sign correction. Thesis chapter 3, equations 3.20–3.22, repeats this structure [2].

For transparent arithmetic choose the illustrative triangular FD

$$u = 60 \text{ km/h}, \qquad w = 20 \text{ km/h}, \qquad k^{\text{jam}} = 150 \text{ veh/km}, \quad k_c = 37.5, \quad C = 2250 \text{ veh/h}. \tag{4}$$

These are review examples, not fitted source-site parameters or a proposed task calibration. The violations below are strict and tolerate small parameter perturbations. One may use $\Delta x = 0.1$ km and $\Delta t = 6$ s, satisfying the source CFL equality. No update trajectory is computed.

2. One lateral departure makes printed receiving negative

Consider one permitted direction from lane l to a free target l' away from route/courtesy effects. Choose the source's allowed congested keep-right setting zero, giving $I = 1$. Other lateral flows are zero. Set

cell position	$i-1$	i	$i+1$	$i+2$
k_l	139	140	141	142
$k_{l'}$	1	2	3	4

All densities are positive and below jam density. At the cells needed here they increase downstream, so the stated source discharge-mode condition $k_i \geq k_{i+1}$ is inactive; the ordinary rules (1)–(2) apply. Since $K_l \geq 140$ and $K_{l'} \leq 4$, the fraction is bounded by

$$\mathcal{P}_{i,l\to l'} \geq \frac{140 - 4}{140 + 4} = \frac{17}{18}, \qquad D_{il} = 2250, \quad S_{il} = 200, \quad S_{il'} = 2250. \tag{5}$$

Hence $L \geq 2125$ and the printed receiving adjustment forces

$$S_{il}^{\text{mod}} = 200 - L \leq -1925, \qquad q_{i-1,l} \leq -1925 \text{ veh/h}. \tag{6}$$

This needs no assumed upstream demand value: the minimum in (3) cannot exceed its negative receiving argument. A departure that should release space instead removes receiving allowance. It is a forward-admissibility obstruction before VSL is added.

3. Two pairwise desires can oversubscribe one donor

Take a three-lane section away from structural incentives, with a middle donor lane $l = 2$. All three lanes are free. At $i, i+1, i+2$, use donor densities $(30, 31, 32)$ and densities $(2, 3, 4)$ on each neighboring lane. The current free-flow keep-right factor, where applied, is $I = 1 - 2/30 = 14/15$; an unrestricted direction has $I = 1$, so

$I \geq 14/15$ for both outgoing directions. Reverse lateral desires are zero. Consequently, independently of the convex lookahead weights,

$$\mathcal{P}_{i,2\to j} \geq \frac{(14/15)30-4}{30+4} = \frac{12}{17}, \qquad j \in \{1,3\}. \tag{7}$$

Both targets have $S = C$; therefore with $D_{i2} = 1800$,

$$L_{i,2\to 1} + L_{i,2\to 3} \geq \frac{24}{17} D_{i2} = \frac{43200}{17} > 1800 \text{ veh/h}. \tag{8}$$

The source density update includes both lateral directions. If equation 20 is extended by summing both, as its conservation update requires for a simultaneous two-direction interpretation, its residual forward demand is

$$D_{i2}^{\text{mod}} = D_{i2} - L_{i,2\to 1} - L_{i,2\to 3} \leq -\frac{12600}{17} < 0. \tag{9}$$

Unlike the first check, this last step explicitly uses the natural summed interpretation of the one-pair printed equation. If instead only one direction is realized, a selection/priority rule must be supplied. The displayed pairwise desires alone do not provide it. At the CFL equality, the requested lateral masses alone total at least 72/17 veh against a current donor sending mass of 3 veh. This is oversubscription of its sending budget, not a claim that a complete unobserved implementation loses mass.

Simply replacing D by a positive VSL-capped budget while retaining the two fractions gives the same ratio $24/17 > 1$. A cap does not allocate the shared budget. Han et al.'s sending restriction [3] therefore cannot alone resolve this issue.

4. Later-source clarification and the decision boundary

Thesis chapter 4, equation 4.11, uses unrestricted flow at $\theta = 0$ and $(1-\theta)\phi$ otherwise. On $0 \leq \theta \leq 1$ this is continuous and gives zero at zero receiving. It differs from the original special branch but neither hand check uses that branch. The later chapter also supplies a three-to-two case. Neither observation resolves the original receiving adjustment or supplies the joint allocator in the inspected summary.

An authoritative algorithm could clarify the transcription or allocation. Absent that evidence, changing signs, clipping or selecting priorities would create a changed flow-realization contract. Qualification stops at that choice. No repaired law, replacement backbone, information advantage, finite-clearance guarantee or derivative contract is selected here.

References

[1] H. H. S. Nagalur Subraveti, V. L. Knoop, and B. van Arem, "First order multi-lane traffic flow model – an incentive based macroscopic model to represent lane change dynamics," *Transportmetrica B*, vol. 7, no. 1, pp. 1758–1779, DOI: 10.1080/21680566.2019.1700846. Source equations 6, 8–9, 13–17, 20–22; repository/copyright 2019, article header 2020.

[2] H. H. S. Nagalur Subraveti, *Lane-specific Traffic Flow Control*, Ph.D. dissertation, TU Delft, 2021, DOI: 10.4233/uuid:e405a549-b1fe-4fa1-8090-4429612359d9. Selected ch. 3, printed pp. 26–32; ch. 4, pp. 51–52.

[3] Y. Han, A. Hegyi, Y. Yuan, S. Hoogendoorn, M. Papageorgiou, and C. Roncoli, "Resolving freeway jam waves by discrete first-order model-based predictive control of variable speed limits," *Transp. Res. C*, vol. 77, pp. 405–420, 2017, DOI: 10.1016/j.trc.2017.02.009, sec. 2.3.